MECHANICAL ADVANTAGE

PYQ • 2021 - 2026

1. 2026 **3 MARKS**

(A) Observe the figures.

(a) Which one has the mechanical advantage less than one ?

(b) State the principle of a lever.

(c) The mechanical advantage of a lever is 2. If the length of its effort arm is 1 m, calculate the length of the load arm.

OR

(B) Crowbar can be used as a lever.

(a) What do you mean by a lever ?

(b) Among Effort, Load, and Fulcrum, state which one comes in the middle for a second order lever and for a third order lever.

(c) A metre scale is suspended in an equilibrium position. If a weight of 60 g wt is suspended at a distance of 25 cm from the balanced point, what weight must be suspended on the other side at a distance of 30 cm to keep the metre scale in equilibrium ?

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2. 2026 **2 MARKS**

A load of 20000 N is lifted with a wheel and axle system that has a wheel of radius 50 cm and an axle of radius 10 cm.

(a) Calculate the mechanical advantage of this wheel and axle.

(b) How should a wheel and axle be designed to obtain increased mechanical advantage ?